



FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

INTEGRATIVE PROGRAMMING AND TECHNOLOGIES III

TEST 2

DATE: 4 MAY 2026

TIME: 45 MINS

EXAMINER: BOAMAH-ABU, CHARLES (PhD)

TOTAL MARKS: 36

NO. OF PAGES: 4 (Including cover page)

INSTRUCTIONS/REQUIREMENTS

1. This examination is closed book
2. Answer all questions
3. Where stated "Briefly answer", please give a short answer
4. Where stated "Write the missing parts" of code, please write only the missing parts
5. Ensure your handwriting is readable

Question 1

10 marks

1. True or false: XML stores data as text. (1)
2. True or false: XML allows you to create your own tags to store data. (1)
1. T; 2. T
3. The table below match the XML technologies to the specific uses. Write **only** the number of a technology and the letter corresponding to the specific use. (4)

	Tools		Used to
1	XSLT	A	Define the structure of XML data
2	XML (Language)	B	Write the XML code to store data
3	XPath	C	Transform how XML data must be displayed
4	XML Schema	D	Navigate to XML nodes or values

Answer: 1. C; 2. B; 3. D; 4. A

4. Briefly explain why is XML is important in Integrative programming? (2)
5. What is the main advantage of Stax over DOM? (2)

Any correct answer, e.g., XML allows different applications to share data; therefore, supports interoperability in Integrative programming

Any correct answer, e.g., compared to DOM, Stax supports larger documents

Question 2

26

The table below shows part of the data on lecturers in the IT Department. Each lecturer has **id**, **name**, **surname** and **phone** number.

1	Sharon	Amp	0612122231
2	Bertha	Boa	0612122232
3	Ngcwele	Mat	0612122233

The data is to be stored as XML in a file called **lecturers.xml**. The root element is **Root**. This contains several **Lecturer** elements. Each **Lecturer** element consists of **id** (an attribute, required) and two elements: **fullname** and **phone**. The **fullname** element consists of **name** and **surname** elements.

1. Complete the XML code (below) for the **first** lecturer in **lecturer.xml** (use **id** as attribute and **Lecturer**, **fullname**, **name**, **surname** and **phone** as tags). (5)

```
<?xml version="1.0" encoding="UTF-8"?>
<Root>
```

```

    _____
    _____
    _____
    _____
    _____
    _____
    _____
    ...

```

```
</Root>
```

Answer:

```
Root>  
  ✓<Lecturer id="1">  
    ✓<fullname>  
      <name>Sharon</name>✓  
      <surname>Amp</surname>✓  
    </fullname>  
    <phone>0612122231</phone> ✓  
  </Lecturer>  
  ...  
  ...  
</Root>
```

2. Write the missing parts of the DTD code in a file **lecturer.dtd**, which defines the structure of the XML in **lecturer.xml**. (9)

```
<?xml version="1.0" encoding="UTF-8"?>  
<!ELEMENT Root (Lecturer)+>  
<!-->  
<!-->  
<!-->  
<!-->  
<!-->  
<!-->
```

Answer

```
<?xml version="1.0" encoding="UTF-8"?>  
<!ELEMENT Root (Lecturer)+>  
<!ELEMENT Lecturer (fullname, phone)> ✓✓  
<!--> ✓✓  
<!--> ✓✓  
<!--> ✓✓  
<!--> ✓  
<!--> ✓  
<!--> ✓
```

3. Link the XML code in **lecturer.xml** to the DTD in **lecturer.dtd**. (2)

```
<!DOCTYPE ____ SYSTEM "____">
```

```
<!DOCTYPE Root ✓ SYSTEM "lecturer.dtd ✓">
```

4. XSL code in a file **lecturer.xsl** defines how the XML in **lecturer.xml** is to be displayed in a web page as shown below.

Id	Name	Surname	Phone
1	Sharon	Amp	0612122231
2	Bertha	Boa	0612122232
3	Ngcwele	Mat	0612122233

Write the missing parts of the XSL code (below) in **lecturer.xsl**, which extracts the data in the XML in **lecturer.xml**. (8)

```
<xsl:template match="/">
...
...
    <xsl:for-each select="_____" />
        < xsl:      =      />
        < xsl:      =      />
        < xsl:      =      />
        < xsl:      =      />
    </xsl:for-each>
...
...
</xsl:template>
```

Answer

```
<xsl:template match="/">
    <xsl:for-each select="Root/Lecturer">✓✓
        xsl:value-of select="@id"/>✓
        <xsl:value-of select="fullname/name"/>✓✓
        <xsl:value-of select="fullname/surname"/>✓✓
        <xsl:value-of select="phone"/> ✓
    </xsl:for-each>
</xsl:template>
```

5. Write the missing parts of the line of code (below) in **lecturer.xml**, which links the XML to the XSL in **lecturer.xsl**. (2)

```
<?xml-stylesheet type="_____" href="_____"?>
```

```
<?xml-stylesheet type="text/xsl"✓ href="lecturer.xsl"✓"?>
```